FarmGuard

A working farming app for Indian smallholders: right crop, healthy field, smart watering, early disease warnings for the whole village, and one-tap messages to family, in Hindi and English.

1. Problem and our answer

Farmer's problem	What FarmGuard does (working today)
Which crop suits my soil and season?	Crop Advisor: soil map + last year's weather + Soil Health Card gives Top-3 crops with yield, profit, reasons and a 3-season rotation
Disease is seen late; no internet in the field	Offline Disease Scan: AI runs on the phone (26 diseases, 14 crops), treatment in Hindi/English
Disease spreads farm to farm	Village Disease Radar: scans from many farms + weather spread risk warn nearby farmers early
Over- or under-watering	Irrigation advice: Irrigate now / Delay / No need, with litres for your land (FAO-56 method)
Can't see crop stress across the field	Satellite Field Health: free Sentinel-2 images give a greenness score, trend and weak-patch warning
Not comfortable with smartphones	Family Helper mode + pre-written messages: tap, the message is written, send on WhatsApp/SMS

2. Our new idea: Village Disease Radar

Every offline leaf scan becomes an anonymous data point (location rounded to ~1 km, no photo, no name). The server groups nearby farms reporting the same disease (DBSCAN clustering, **one vote per farm**), checks if the next 3 days' weather favours its spread (e.g. **Hutton criteria** for late blight), and warns only farmers **growing that crop** nearby, plus their family helpers, each in their own language. A diagnosis tool becomes an early-warning network for the village.

3. Features (all built, tested, Hindi + English)

Feature	What judges will see	Mode
Offline Disease Scan	Leaf photo, result in under 1 s with internet OFF; rejects non-leaf photos; treatment guide; history; share	**FALLBACK** on-device
Village Disease Radar	Map of anonymous cases + outbreak circles; Watch / Warning / Outbreak; alerts; DEMO mode clearly labelled	**LIVE** weather / **FALLBACK** saved
Smart Crop Advisor	4-step wizard; Top-3 crops, % match, yield, profit/acre, sowing window, reasons, soil tips, rotation plan	**LIVE** SoilGrids + archive
Soil Health Card	AI reads the card photo, or the farmer types 5 numbers	**LIVE** AI / **FALLBACK** manual
Irrigation + Weather	7-day forecast, heavy-rain/heat/wind warnings, irrigation decision with reasons and litres	**LIVE** Open-Meteo / **FALLBACK** saved
Satellite Field Health	Score 0-100, NDVI/NDMI, 6-month chart, 'dropping' / 'patchy' / 'dry' warnings	**LIVE** Sentinel-2 / **FALLBACK** estimate
Family Helper	6-digit link code, helper acts for farmer, view/edit permission, Help-me button	always
Quick Messages	Tap template, blanks from dropdowns, auto-written in contact's language, WhatsApp / SMS	always (offline)
Farming Assistant	Chat in Hindi/English from the farmer's own data + built-in FAQ; AI when a key is added	**LIVE** AI / **FALLBACK** FAQ
Schemes, Rent and Waste	8 government schemes with official links; rent tractors and sell straw nearby via WhatsApp	always
Installable PWA	Installs on a phone and opens with no internet (service worker)	offline

4. LIVE / FALLBACK: never breaks, never fakes

Every feature works with an online service (LIVE) and without one (FALLBACK: last saved forecast, built-in data, on-device AI). Each result shows a badge with its source; a System Status page lists all services. AI use has a daily limit to protect credits. Demo data is always labelled.

5. Proof it works

Check	Result
Automated tests	83 backend (pytest) + 18 frontend (vitest), all passing; Hindi/English completeness tested
Offline disease model	7/7 test photos correct in a real browser, also with all model files blocked (served from phone cache)
Radar (real weather)	7 demo farms near Ludhiana gave OUTBREAK; Hindi alert to farmer, English to neighbour and helper
Crop Advisor (real data)	Ludhiana, after rice: Rabi gave chickpea / mustard / wheat; Kharif gave cotton / pigeon pea / groundnut
Satellite (real data)	Rice field greenness 0.12 (June) to 0.45 (Sept) to 0.27 (Oct harvest), matching the real crop cycle
Offline app (PWA)	App opens with the network switched off after the first visit

6. Technology (free and open)

React + Vite + Tailwind + Leaflet (PWA) | FastAPI + SQLite | ONNX Runtime Web (DINOv2-small, CC BY-SA 4.0) | scikit-learn DBSCAN | Open-Meteo, ISRIC SoilGrids, Sentinel-2 via Microsoft Planetary Computer | any OpenAI-compatible AI (optional). No paid API is required.

7. Honest limitations

• Disease model knows 14 crops (PlantVillage); field photos score lower than lab accuracy (98.7%). No rice/wheat diseases yet.
• Radar needs many users in an area; until then weather-only disease risk works for every farmer.
• Soil maps are 250 m estimates; satellite analyses a ~330 m box, not exact field boundaries; monsoon clouds block images.
• Prices/yields are estimates (Kharif MSPs to be re-verified). WhatsApp messages are pre-filled; the user presses Send.

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